

VRisbee: How Hand Visibility Impacts Throwing Accuracy and Experience in Virtual Reality

Malte Borgwardt
borgward@uni-bremen.de
University of Bremen
Bremen, Bremen, Germany

Jonas Boueke
jboueke@uni-bremen.de
University of Bremen
Bremen, Bremen, Germany

María Fernanda Sanabria
mariafer@uni-bremen.de
University of Bremen
Bremen, Bremen, Germany

Michael Bonfert
University of Bremen, Digital Media
Lab
Bremen, Bremen, Germany
bonfert@uni-bremen.de

Robert Porzel
University of Bremen, Digital Media
Lab
Bremen, Bremen, Germany
porzel@uni-bremen.de



Figure 1: Scene of the virtual reality Frisbee simulation developed for the study

ABSTRACT

Hand interaction plays a key role in virtual reality (VR) sports. While in reality, athletes mostly rely on haptic perception when holding and throwing objects, these sensational cues can be missing or differ in virtual environments. In this work, we investigated how the visibility of a virtual hand can support players when throwing and what impact it has on the overall experience. We developed a Frisbee simulation in VR and asked 29 study participants to hit a target. We measured the throwing accuracy and self-reports of presence, disc control, and body ownership. The results show a subtle advantage of hand visibility in terms of accuracy. Visible hands further improved the subjective impression of realism, body ownership and subjective control over the disc.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Virtual reality**.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585868>

KEYWORDS

virtual reality, throwing, embodiment, self-avatar, sports

ACM Reference Format:

Malte Borgwardt, Jonas Boueke, María Fernanda Sanabria, Michael Bonfert, and Robert Porzel. 2023. VRisbee: How Hand Visibility Impacts Throwing Accuracy and Experience in Virtual Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585868>

1 INTRODUCTION

Every sport requires specific technical abilities to achieve the main objective, for example, to throw a Frisbee accurately is required for *Ultimate* or *Disc Golf*. Frisbee throwing skills and tactics differ from other sports, as throwing a disc relies on other physical principles than throwing a ball or a dart, for example. A disc travels on a flat trajectory stabilized by spin and lift forces that are highly dependent on the disc's orientation. Since virtual reality (VR) technology evolved and gained popularity, it can help improve or practise skills that can be more difficult to replicate in a real-world setting [21]. Approaches range from learning musical instruments, training astronauts, and educating in the military or medical fields, to applications in sports. For skill acquisition in virtual simulations, the learned skill needs to be transferable to reality, e.g., for throwing real darts [16, 21]. However, especially

for throwing, the transfer between the virtual and the real world comes with challenges and currently requires trade-offs, e.g., in terms of distance perception [4] or aiming [23].

Throwing a disc additionally requires players to consider the weight and shape of the disc as well as external factors such as wind through haptic stimuli to anticipate the possible outcome of the throw. In current consumer VR systems with controllers or hand tracking, these cues are not available to the player. How can we still enable players to accurately throw a disc and control its flight?

We investigated supporting Frisbee throwing with a visual hand representation. In VR setups, users are often supported with visual self-representations such as full or partial self-avatars. Studies found significant benefits of hand visibility for tasks like typing on a virtual keyboard [8] or picking and placing objects [1, 5] as it has a positive influence on players' presence and body ownership in the virtual world. This study aims to understand if hand visibility has an impact on the performance and sense of embodiment when throwing a disc in VR. Therefore, we investigate the following hypotheses: **H1** The visibility of the virtual hands improves the accuracy of throwing a virtual Frisbee; **H2** Hand visibility increases perceived presence; **H3** Hand visibility increases the sense of body ownership; **H4** Hand visibility increases the subjective control over the virtual Frisbee. We report on the design and results of a user study ($N = 29$) and discuss the outcome. Further, we contribute a prototype with realistic Frisbee trajectory physics.

2 RELATED WORK

Different research areas are related to this interdisciplinary topic, such as human-computer interaction, sports, computer science, and psychology. We present related work concerning aspects of embodiment and its effects on interactions and sports in VR.

2.1 Aspects of Embodiment

Kilteni et al. [12] explain the sense of embodiment as a combination of three concepts: self-location, sense of agency, and sense of body ownership. For a plausible sense of self-location, it is important that the biological and virtual bodies are perceived to be in the same space [5]. A high sense of agency requires having motor control over the virtual body's actions [12]. The sense of body ownership refers to "one's self-attribution of a body" [12] and implies that the body is the agent of the experienced sensations [7]. The user's perceived level of embodiment can strongly affect the VR experience. The concept is linked to the user's presence in a virtual environment [18, 20] and might therefore determine how well experiences and skill acquisition from the virtual world can be transferred to the real world.

There is a growing body of literature on how VR training in sports can benefit performance in reality. In a study by Pastel et al. [19], participants completed three tasks in both virtual and real environments: balance, grasping and throwing. Its results showed that full body visualisation led to better performances, especially if compared with no body visualisation at all, and that a realistic virtual body helps to "limit differences between the real world and virtual reality and to ensure quite natural body perception" [19]. Research on body ownership concerning the user's virtual hands by Canales et al. [5] investigated the influence of different mappings

on a manipulation task in virtual reality. They found that participants preferred adjusted visualisations that prevented the hand from intersecting with objects, even though displaying the original position improved performance, which also induced higher ownership. Similarly, Ma and Hommel [15] showed the role of agency for perceived hand ownership. The degree to which the virtual agent could be controlled by people's own movements was at least as important as a realistic appearance of the hand when it comes to body ownership. Further, a study by Grubert et al. [8] evaluated different hand and fingertip representations to find the best visual support for extensive text entry. Minimalist visualisation of only the fingertips proved to be the preferred option, with a high input rate and low error rate. This outperformed the high-fidelity variant of a full 3D representation of the hand, which in turn induced higher body presence at the cost of obscuring the keyboard. Therefore, it might not be possible to generalize the ideal hand representation for any use case, and rather depend on the scenario, the goal, the input and output devices, and individual preferences. This is also supported by the findings of Kocur et al. [13], who conducted a study to investigate the impact of missing individual fingers in virtual interactive scenarios. They found, that a missing virtual representation of the index finger, which was by far the most used by the participants, led to the strongest drop in body presence. Comparing this to the effect of replacing the virtual hand by an abstract one, it showed, that the usefulness of a hand representation has a much higher influence than realism.

2.2 Sports in Virtual Environments

With innovations in sensor technology enabling position, motion and biofeedback detection for complex posture recognition or physiological measurements, doing sports in virtual environments opens up new opportunities. Some systems are particularly designed for coaching and training people, both for supporting physical activity and providing motivation [2].

Neumann et al. [17] concluded in their systematic review of interactive sports applications in VR that while "the majority of research has been conducted on endurance sports, such as running, cycling, and rowing, more research is required to examine the use of interactive VR in skill-based sports" [17]. This is particularly challenging as precise movements and actions, such as throwing, are difficult to capture and transfer accurately to virtual actions. For example, Harris et al. [9] investigated the performance of golf putting in VR and compared it to the performances in the real world. Their study showed that expert players also outperformed novices in the virtual world, suggesting that a VR simulation can achieve a high level of fidelity so that it requires skills comparable to the real world. As another example, a study by Tirp et al. [21] on skill transfer of throwing darts compared the accuracy and quiet eye duration between real and virtual throwing. The results showed that participants with virtual training threw with the highest accuracy which implies a high skill transferability. However, these results conflict with findings by Drew et al. [6]. In a similar study, the participants who practised dart throwing in VR performed worse than the real-life training group. Furthermore, a discrepancy between real and virtual throwing is demonstrated in an investigation by Zindulka et al. [23] who compared overhand and underhand ball

throws. Their results showed a significantly worse performance in VR compared to real throwing regarding accuracy and precision, with pronounced scattering especially in the vertical axis. These findings could be explained by using controllers for throwing. Timing the release of the object with the controller's trigger button seems challenging compared to releasing a physical ball from the fingers. Only small deviations from the optimal release moment can lead to a strong impact on the trajectory of the virtual object. Although a disc with a horizontal trajectory is thrown in our study, the limited control over the release point might also be a limiting factor of our controller-based system.

Concerning the embodiment during sports in VR, Kosmalla et al. [14] tested the influence of limb visualisation on the climbing experience in VR. The climbers' hands and feet were tracked and the virtual route matched the physical wall. The authors reported that the perceived stepping errors were significantly higher without foot visualisation. However, the perceived gripping errors were not affected by showing or hiding the virtual hands. With more experience in climbing, the participants were able to compensate for the lack of hand visibility by only relying on proprioceptive cues. Similarly, Bonfert et al. [3] analysed the influence of foot visibility on a kicking task in VR. They found that the visible foot improved the accuracy of the shot, even though the subjective control over the virtual ball did not improve. The self-reported body ownership was reported higher with the visible foot. While these studies show that the visibility of the hands and especially feet can improve the performance and user experience in VR sports, they do not address the influence of hand visibility in throwing tasks. We are trying to fill this gap with our study using the system we specifically developed for this purpose.

3 SYSTEM DESIGN

To learn about the influence of hand visibility in VR sports, we created a VR game that simulates the throwing of a Frisbee. It was built for the Meta Quest 1 with its standard Touch Controllers. The object interaction was based on the physics engine and the XR Interaction Toolkit of Unity 2021.3.3f1. The application allowed grasping discs and launching them with the speed and direction of the controller motion. However, as the disc trajectory is not parabolic, the hovering and curve are not calculated realistically by the physics engine and had to be addressed manually. The implementation of the Frisbee flying physics is grounded in the works by Hubbard and Hummel [10] and Hummel [11] who developed a numerical simulation and estimated the model's aerodynamic coefficients to match the trajectory of a real disc. We adjusted the virtual disc's parameters of translational velocity, angular velocity, and tilt in Unity according to the estimated forces of the initial impulse, lift, drag, and torques. One part of the calculation addresses the disc's lift (L) and drag (D), they are calculated as shown in Equation 1. Both are dependent on the plan-form area of the disc (A), the velocity (v), the air density (ρ) as well as a lift coefficient (C_l) for lift and a drag coefficient (C_d) for drag. C_l is linearly dependent on α while C_d is dependent on α^2 [10].

$$L = \frac{1}{2} C_l A \rho v^2, \quad D = \frac{1}{2} C_d A \rho v^2 \quad (1)$$

For further details on Frisbee physics, we refer to the papers by Hubbard and Hummel [10] and Hummel [11], and for the actual implementation to our source code¹.

The game's virtual environment consists of a table in front of the player on which the Frisbee spawns after each throw as well as a board with information about the score, number of throws, misses and time tracking. 10 meters from the player is a round bullseye target with a diameter of 3 meters (see Figure 1). When a disc hits the target, it stays at the collision point as a reference, and the score is increased. The target is segmented into five rings with different score values assigned – the closer to the centre, the higher the score – to motivate the player to aim at the centre. For every throw, the disc's distance to the centre of the target was logged. In case it missed the target, we logged the distance and direction from the disc to the point where it intersected a plane extending the target's surface, or where it hit the ground if it did not pass the target. This way, we not only tracked if the target was missed but also by how far and where it was missed.

For the visualisation of the player's hands in one condition, we implemented floating hands that were animated to a gripping pose when picking up a disc and opened up again when releasing it. Participants in the other condition did not see their virtual hands. In both conditions, a white beam appears when pointing at the disc to indicate that it can be picked up. When the user then presses the hand trigger of the Touch Controller that lies under the middle finger, the disc is moved towards the hand and follows it. The interaction is displayed in Figure 2.

4 STUDY

We conducted a user study ($N = 29$) to observe the effect of hand visibility on the accuracy, presence, perceived body ownership and agency when throwing a Frisbee. Participants were asked to pick up and throw a disc onto a circular target with the goal of hitting as close to the centre as possible. We performed a double-blind study in which neither the participants nor the instructors of the study knew which condition had been selected for the specific trial at hand. The selection of the conditions of *no hands* or *hands* was counter-balanced across the sample. To avoid learning effects as a confounding factor on performance and habituation effects from the embodiment, we chose a between-group design, thus participants played the game only in one of the two conditions.

In total 29 people participated in the study: 15 without visible hands and 14 with visible hands. The age ranged from 20 to 71 with a mean of 33 ($SD = 14, 2$). In total, 16 participants identified as female and 13 as male. The condition without hands was tested by 6 female and 9 male participants, and the condition with hands by 10 female and 4 male participants. In both groups a small proportion of the participants was left-handed (2 out of 15 (13%) for the no hands condition and 3 out of 14 (21%) for the with hands condition), while the rest threw with their right hand. The self-assessment of how athletic they are and their proficiency in throwing a Frisbee also turned out to be balanced between the groups. Three of the participants had extensive prior experience in VR (2 in the group without hands, and 1 in the group with hands).

¹<https://github.com/MalteBorgwardt/VRisbee>

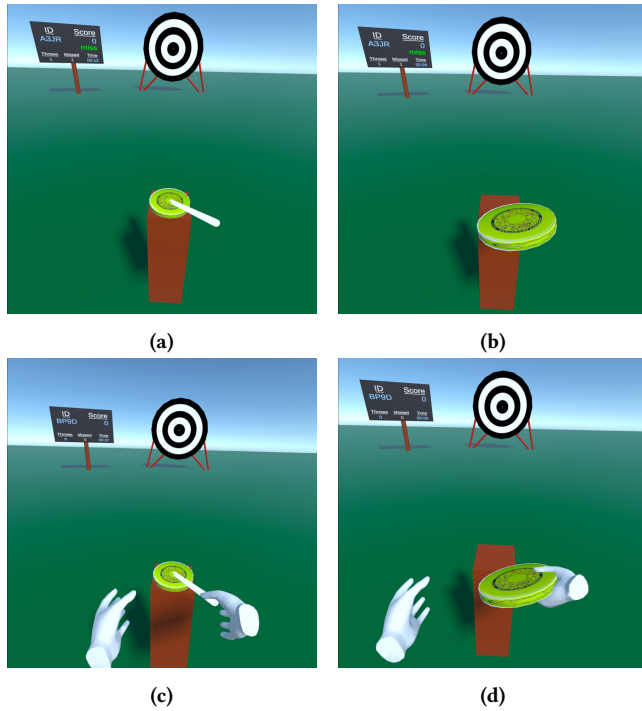


Figure 2: The hand visualisations of the two conditions before and after grasping the Frisbee. From left to right: (a) Invisible hands before grasping the disc, (b) invisible hands holding the disc, (c) visible hands before grasping the disc, (d) visible hands holding the disc.

The study took place indoors in a spacious and closed room. Every player had 10 training shots and, subsequently, 30 trials to score as many points as possible in a standing position. The duration was unlimited and all participants were told explicitly they had as much time as they wanted or needed for their throws. The average time taken by the participants for their 30 trials was 3 minutes and 37 seconds. The whole experiment took on average approximately 40 minutes with the introduction, test trials, experimental trials, questionnaire, optional trials for fun, and interview. During the game, metrics for each throw were logged and stored on the VR headset after the experiment was finished. After completion, the participants filled in the following questionnaires on a laptop – all questions were shown in English with a German translation provided:

- Demographic questions: age, gender, height, handedness (left or right)
- Prior experience: athleticism in general, proficiency in throwing a Frisbee, experience with VR headsets
- Presence Questionnaire by Witmer & Singer [22] excluding the sound and haptic components
- Custom questions on body ownership as listed in Table 1.
- Custom questions on the control over the Frisbee as listed in Table 1.

Concluding the experiment, each of the participants was asked for feedback about their perception of the game and its mechanics.

Body ownership

- I was aware of the location of my biological hands.
- I was aware of the location of my virtual hands.
- I felt that my biological hands were present in the virtual environment.
- I knew exactly where my hands were in the virtual environment.
- I was confused about my hands in the virtual environment. (reversed)

Control over the disc

- How precisely were you able to control the direction of the throw?
- Did the disc hit the spot you aimed for?
- Did the disc behave the way you wanted it to?
- How satisfied were you with the result you achieved?
- How realistic was the throw?

Table 1: Custom questions of the questionnaire on body ownership and control over the Frisbee.

5 RESULTS

With the collected data we tested our hypotheses. We assume that hand visualisation has advantages, and therefore performed one-sided tests. In all our tests we used an alpha level of .05.

5.1 Accuracy

Since the data on **throwing accuracy** is not normally distributed according to the Shapiro-Wilk-Test ($p < .001$), we applied a Mann-Whitney U test. If we take all throws into account, we observe a significant difference ($p = .008$) with a negligible effect size of $r_{tb} = .097$. The mean accuracy was a distance to the centre of 1.672 m for the with hands condition and 1.958 m for the no hands condition. When looking solely at the throws not hitting the ground, the difference is not significant ($p = .096$). The respective means are 1.446 m for with hands and 1.474 m for no hands. This means, throws in the with hands condition were generally closer to the centre and therefore more likely to hit the target. When looking at the results of left-handed and right-handed participants separately, the scatter plots show an interesting characteristic, as illustrated in Figure 4. The hits and misses of right-handed throws tend to be deflected around an axis from bottom-left to top-right. For left-handed throws, the deflection tends to be from bottom-right to top-left, instead.

5.2 Questionnaires

To evaluate each subscale of the Presence Questionnaire by Witmer & Singer [22] and the two custom questionnaires on body ownership and control over the disc, we used Mann-Whitney U tests.

From the **Presence** Questionnaire, only the category ‘realism’ showed a significant difference ($p = .002$) with a large effect size ($r_{tb} = -.654$) in favor of the with hands condition. Box plots for ‘realism’ are shown in Figure 3. For the other categories ‘possibility to act’, ‘quality of interface’, ‘possibility to examine’, and ‘self-evaluation of performance’ there was no significant difference with p -values $p > .161$ for each category.

For the overall **body ownership**, the difference was not significant ($p = .051$). As this test is close to a significant result, it is conceivable that with a larger sample, a possible effect might be

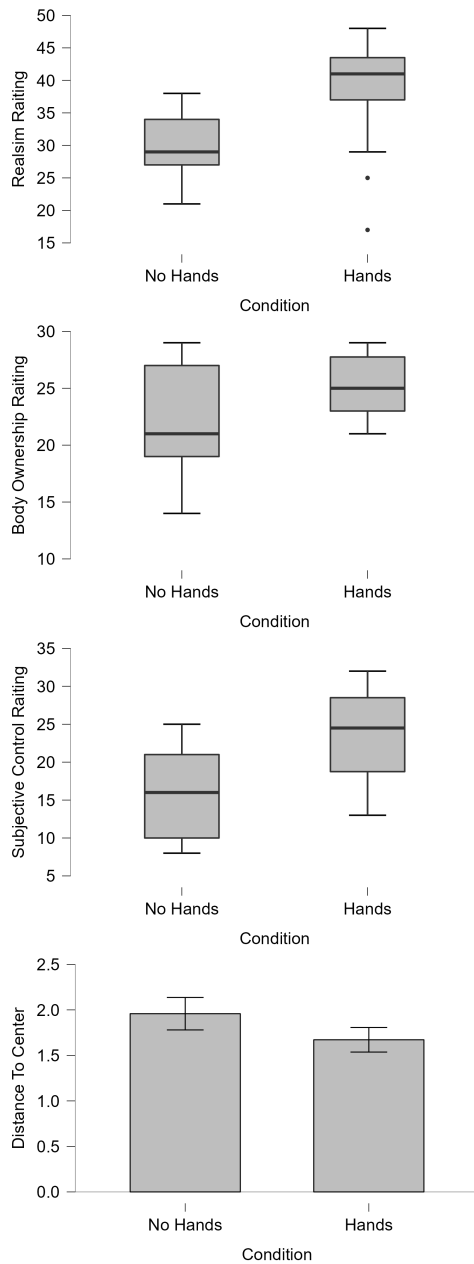


Figure 3: Box plots for the rating of realism, body ownership and subjective control and a bar plot for the accuracy with a 95% confidence interval for the two conditions no hands and hands.

measurable. In the distributions of the data visualised in Figure 3, we can see the distributions are close to each other but also have some differences. There is a large range of opinions for the without hands condition in contrast to the with hands condition.

The **subjective control** items show a significant difference ($p = .004$) with a large effect size of $r_{th} = -.599$. The box plots for the data on subjective control are shown in Figure 3.

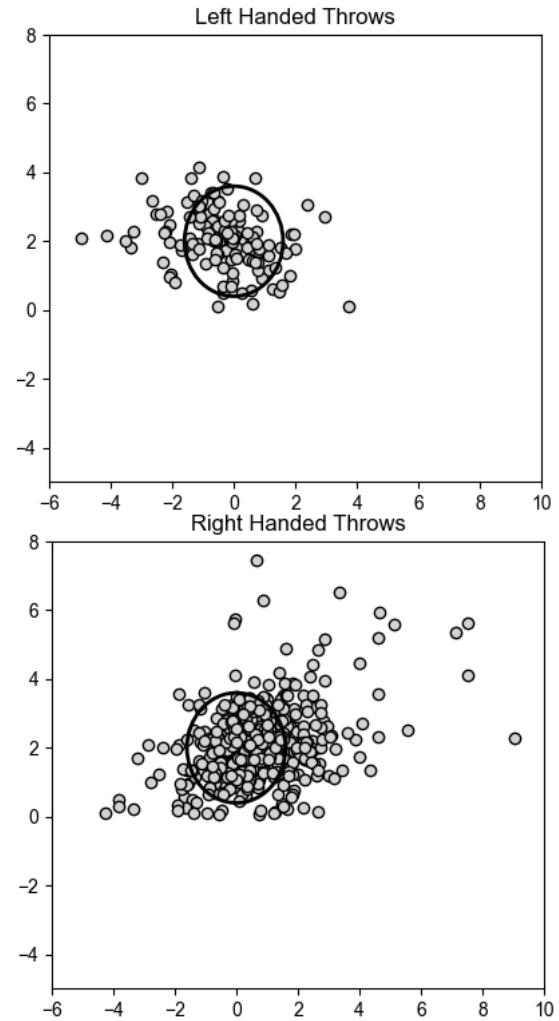


Figure 4: Scatter plots of the dispersion pattern with circles showing the actually circular target, top: left-handed throws, bottom: right-handed throws. The axis are displayed in meters.

5.3 User Feedback

The overall feedback of the experiment was that the participants had fun. Many wanted to continue playing after the 30 throws and made 20-70 more throws. Participants reported that the controllers were not perceived as negatively influencing the throw, but three participants used the controller in different ways for picking up and throwing the virtual disc.

6 DISCUSSION

Our results show that hand visibility improves accuracy significantly but with a small effect size. Both groups have a similar data distribution, and the mean accuracy of the with-hands condition was slightly better. This only cautiously supports hypothesis H1 as the throwing accuracy does not strongly depend on whether the hands are visible or not. Since all participants interacted with the

system for the first time and throwing in virtual reality generally is a challenging task, the accuracy of the throws might have suffered from unusual usage of the controllers. Still, our sample covered a wide range of ages and expertise in VR or sports, thus the result can be seen as representative.

A second look into the dispersion pattern shows two things. First, the scattering tends to be more prominent in the horizontal axis, which is in line with the findings of Zindulka et al. [23] as described in section 2, where overhand and underhand throws resulted in larger scattering around the vertical axis. Second, left-handed throwers would rather hit the left side of the target while the hits of right-handed throwers aimed more to the right. Given the movement of a natural backhand throw of a disc, this might indicate that the release timing of the disc with a controller is difficult and players tend to release the disc too late.

Regarding H2, we saw, that visible hands only had a positive influence on the presence subscale *Realism*. While visible hands induced a more realistic scenario, we could not identify other impacts on the feeling of “being there” in the virtual world, and being able to interact with it. This could be due to the fact, that even with invisible hands, participants were able to manipulate and throw the disc as they wanted to.

The results on body ownership showed no significant difference (H3) but are borderline. Therefore, we cannot assess if body ownership is affected by hand visibility or not. It would seem plausible considering the accuracy of the throws. With only a small impact through visible hands on the accuracy, this indicates that the visual focus while aiming lies mainly on the target and the launch is led by proprioception or muscle memory rather than visual cues.

In addition, with visible hands, the subjective control was rated significantly higher. The participants with visible hands had the feeling of having more control and being able to throw the disc more accurately (H4). This result could also be related to the fact that the system required the players to pick up a new disc manually, which could be a more difficult task with invisible hands, and therefore influence the feeling of control negatively.

6.1 Limitations

These findings are the subjective perception of the participants on their experience of the whole game. That includes not only the throw itself but also the mechanism of picking up a new disc to throw next. Here, players with a visible hand could have an advantage in hand-eye coordination, leading to a quicker feeling of success compared to the group with invisible hands. It is also important to mention that the disc does not perfectly behave the way it would in the real world. For example, the curve behaviour of the disc when it is thrown with an angle is not as prominent as with real discs, which could irritate participants with experience in throwing a disc.

6.2 Future Work

Future research in the area may include a wide range of options for throwing a disc in VR that can also help understand hand or body visibility. Those could be, for example, hand tracking for bare-hand interaction with no haptic feedback at all instead of using a controller. Alternatively, the Valve Index Controller could be

used in this type of research as it allows tracking the position and pressure of every finger as well as releasing the grasp around the controller handle for releasing the held object. This might lead to a more realistic experience when grabbing and releasing a disc, and possibly a better outcome in terms of accuracy and precision. Another step towards realistic sensations can be haptic feedback gloves, which give users force feedback on all fingers, in addition to precise hand and finger tracking. Nevertheless, the weight and centre of mass of the virtual disc would not be rendered.

Further studies may also focus not only on the first few sets of throws but focus on long-term practice. This way, the influence of hand visibility can be investigated in more detail for experienced players as its effect might decrease with a more reliable muscle memory over time.

7 CONCLUSION

In this study, we analysed the influence of hand visibility on Frisbee throws. We found, that seeing virtual hands does not improve the accuracy of the throws considerably. Nonetheless, the subjective control over the disc was significantly higher in the group with visible hands. It seems, that visible hands raise the feeling of more control in the virtual world and support the user in secondary tasks such as picking up the disc before throwing it. Still, we could see similar results as Zindulka et al. [23] for throwing a virtual disc, in essence, that the release timing in general is a difficult challenge and has a strong impact on where the disc lands.

We also found that hand visibility has a positive significant influence on perceived realism, while the overall perceived presence did not benefit from visible hands. Also, we could not confirm that hand visibility has a significant positive influence on body ownership in disk throwing but inconclusive results. In contrast to the studies on kicking in VR by Bonfert et al. [3] and climbing in VR by Kosmalla et al. [14], our findings showed that a visual representation of the limbs alone does not suffice for a throwing task, which requires highly adjusted motor skills while already holding the virtual object.

ACKNOWLEDGMENTS

This work was partially funded by the Klaus Tschira Foundation, the KI-SIGS Project “AP 390” of the German Federal Ministry of Education and Research (BMBF), and the German Research Foundation (DFG) as part of the Collaborative Research Center (SFB) 1320 EASE – *Everyday Activity Science and Engineering*, University of Bremen (www.easeccr.org).

REFERENCES

- [1] F. Argelaguet, L. Hoyet, M. Trico, and A. Lecuyer. 2016. The role of interaction in virtual embodiment: Effects of the virtual hand representation. In *2016 IEEE Virtual Reality (VR)*. 3–10. <https://doi.org/10.1109/VR.2016.7504682>
- [2] Arnold Baca, Peter Dabnichki, Mario Heller, and Philipp Kornfeind. 2009. Ubiquitous computing in sports: A review and analysis. *Journal of sports sciences* 27 (09 2009), 1335–46. <https://doi.org/10.1080/02640410903277427>
- [3] Michael Bonfert, Stella Lemke, Robert Porzel, and Rainer Malaka. 2022. Kicking in Virtual Reality: The Influence of Foot Visibility on the Shooting Experience and Accuracy. In *Proceedings of the 2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE Computer Society, Virtual. <https://doi.org/10.1109/VR51125.2022.00092>
- [4] Karolis Butkus and Tautvydas Ceponis. 2019. Accuracy of throwing distance perception in virtual reality. In *IVUS*.

- [5] Ryan Canales, Aline Normoyle, Yu Sun, Yuting Ye, Massimiliano Di Luca, and Sophie Jörg. 2019. Virtual Grasping Feedback and Virtual Hand Ownership. In *ACM Symposium on Applied Perception 2019* (Barcelona, Spain) (SAP '19). Association for Computing Machinery, New York, NY, USA, Article 4, 9 pages. <https://doi.org/10.1145/3343036.3343132>
- [6] Stefanie A. Drew, Madeline F. Awad, Jazlyn A. Armendariz, Bar Gabay, Isaiah J. Lachica, and Jacob W. Hinkel-Lipsker. 2020. The Trade-Off of Virtual Reality Training for Dart Throwing: A Facilitation of Perceptual-Motor Learning With a Detriment to Performance. *Frontiers in Sports and Active Living* 2 (2020). <https://doi.org/10.3389/fspor.2020.00059>
- [7] Shaun Gallagher. 2000. Gallagher, S. 2000. Philosophical Conceptions of the Self: Implications for Cognitive Science. *Trends in cognitive sciences* 4 (02 2000), 14–21. [https://doi.org/10.1016/S1364-6613\(99\)01417-5](https://doi.org/10.1016/S1364-6613(99)01417-5)
- [8] Jens Grubert, Lukas Witzani, Eyal Ofek, Michel Pahud, Matthias Kranz, and Per Kristensson. 2018. Effects of Hand Representations for Typing in Virtual Reality. 151–158. <https://doi.org/10.1109/VR.2018.8446250>
- [9] David J Harris, Gavin Buckingham, Mark R Wilson, Jack Brookes, Faisal Mushtaq, Mark Mon-Williams, and Samuel J Vine. 2020. Exploring sensorimotor performance and user experience within a virtual reality golf putting simulator. *Virtual Reality* (2020), 1–8. Publisher: Springer.
- [10] Mont Hubbard and S. Hummel. 2000. Simulation of Frisbee Flight. (01 2000).
- [11] Sarah Ann Hummel. 2003. Frisbee Flight Simulation and Throw Biomechanics. (2003). <https://morleyfielddgc.files.wordpress.com/2009/04/hummelthesis.pdf> last access: 02.08.2022, Master Thesis, University of California, Davis.
- [12] Konstantina Kiltani, Raphaela Groten, and Mel Slater. 2012. The Sense of Embodiment in Virtual Reality. *Presence Teleoperators & Virtual Environments* 21 (11 2012). https://doi.org/10.1162/PRES_a_00124
- [13] Martin Kocur, Sarah Graf, and Valentin Schwind. 2020. The Impact of Missing Fingers in Virtual Reality. In *Proceedings of the 26th ACM Symposium on Virtual Reality Software and Technology* (Virtual Event, Canada) (VRST '20). Association for Computing Machinery, New York, NY, USA, Article 4, 5 pages. <https://doi.org/10.1145/3385956.3418973>
- [14] Felix Kosmalla, André Zenner, Corinna Tasch, Florian Daiber, and Antonio Krüger. 2020. The Importance of Virtual Hands and Feet for Virtual Reality Climbing. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3383067> event-place: Honolulu, HI, USA.
- [15] Ke Ma and Bernhard Hommel. 2015. The role of agency for perceived ownership in the virtual hand illusion. *Consciousness and Cognition* 36 (2015), 277–288. <https://doi.org/10.1016/j.concog.2015.07.008>
- [16] Seyyed Ahmad Mousavi, Mehdi Shahbazi, Elahe Arabameri, and Elham Shirzad. 2018. The Effect of Virtual Reality Training on Learning and Kinematics Characteristics of Dart Throwing. *International Journal of School Health* (2018).
- [17] David Neumann, Robyn Moffitt, Patrick Thomas, Kylie Loveday, David Watling, Chantal Lombard, Simona Antonova, and Michael Tremmer. 2018. A systematic review of the application of interactive virtual reality to sport. *Virtual Reality* 22 (09 2018). <https://doi.org/10.1007/s10055-017-0320-5>
- [18] Nicolas Nostadt, David A. Abbink, Oliver Christ, and Philipp Beckerle. 2020. Embodiment, Presence, and Their Intersections: Teleoperation and Beyond. *J. Hum.-Robot Interact.* 9, 4, Article 28 (may 2020), 19 pages. <https://doi.org/10.1145/3389210>
- [19] Stefan Pastel, Chien-Hsi Chen, Katharina Petri, and Kerstin Witte. 2020. Effects of body visualization on performance in head-mounted display virtual reality. <http://dx.doi.org/10.25673/36147>
- [20] Richard Skarbez, Frederick P. Brooks, Jr., and Mary C. Whitton. 2017. A Survey of Presence and Related Concepts. *Comput. Surveys* 50, 6 (Nov. 2017), 96:1–96:39. <https://doi.org/10.1145/3134301>
- [21] Judith Tirp, Christina Steingröver, Nick Wattie, Joe Baker, and Jörg Schorer. 2015. Virtual realities as optimal learning environments in sport – A transfer study of virtual and real dart throwing. *Psychological Test and Assessment Modeling* 57 (01 2015), 57–69.
- [22] Bob G. Witmer and Michael J. Singer. 1998. Measuring Presence in Virtual Environments: A Presence Questionnaire. *Presence: Teleoperators and Virtual Environments* 7, 3 (06 1998), 225–240. <https://doi.org/10.1162/105474698565686> arXiv:<https://direct.mit.edu/pvar/article-pdf/7/3/225/1836425/105474698565686.pdf>
- [23] Tim Zindulka, Miroslav Bachinski, and Jörg Müller. 2020. Performance and Experience of Throwing in Virtual Reality. 1–8. <https://doi.org/10.1145/3313831.3376639>